Data-driven regional traffic simulation from heterogeneous mobility sources

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Abstract

Knowing how much traffic each road carries requires combining data sources that each see only part of the picture. We built an agent-based simulator of hourly traffic on the road network of Emilia-Romagna, Italy, that combines three of them: mobile-phone data provide the demand and its daily rhythm, loop detectors set its level, and GPS traces guide the choice of routes. We then compared the simulator with data that did not set its values, including detectors in a later season, and with simple predictors without simulation. The simulator reproduces the overall traffic volume, the ranking of roads and the daily timing. At a single road, the hourly error remains large, and simple predictors perform similarly. We also measured what each source adds: detectors fix the level, GPS improves the ranking of roads, and phone data give the timing and the typical trip length, while specific origin–destination pairs add no detectable value. The simulator offers a basis for future scenario studies.

Keywords: traffic reconstruction, mobile-phone demand, inductive loop detectors, GPS probes, route choice, agent-based simulation

1 Introduction

Transport planning [1], the study of congestion [2] and the assessment of emissions [3] all need to know how much traffic each road carries, hour by hour. No single data source provides this. Mobile phones show where people travel but not which roads they use, and phone samples have their own biases [4–6]. Loop detectors count every vehicle, but only at a few points, and where they are placed limits what we can learn about the rest of the network [7–10]. GPS-equipped vehicles cover many roads, but they are a small sample of traffic, and the chance of sampling a vehicle changes between roads and hours [11–13]. A regional model therefore has to combine these sources [14–17]. This raises a simple question: what does each of them actually add?

Each source has been studied for a long time. Phone and GPS data have shown statistical regularities in how individuals move [18–23]. Phone records have also been used to estimate origin–destination (OD) matrices [24, 25], and these matrices have been assigned to road networks [26] to study road use and congestion [1, 2, 27]. Detector counts have long been used to estimate OD matrices [28–30] and to calibrate agent-based simulations [31]. GPS probes and fixed sensors are used to estimate the state of traffic, mostly on motorways and urban corridors [32–37].

More recent work combines the sources. Telecom data and vehicle flows have been used together to predict traffic where there is no camera [38], traffic has been estimated at places without sensors [39], and a few fixed sensors have been scaled up to describe whole urban networks [40]. Closer to our work, agent-based simulations now take their demand from phone data. In Barcelona, activity diaries built from phone records feed a MATSim model [41]. In Novi Sad, telecom data are combined with population and GIS data, and traffic is calibrated and validated with vehicle counters [3]. Aggregated phone data have also been used to build a

model of Brussels [42]. A recent review of more than 60 such case studies shows how common these data-driven, city-wide simulations have become [43]. Combining the sources is therefore no longer new, but doing so does not by itself show what each of them adds.

Three things make this hard to see. First, independent data are scarce. Simulations should be validated on data not used for calibration [44, 45], yet demand is often rescaled to match the counts, and agreement at those same counts is then reported. The closest studies show how few options there can be: Çolak et al. report that no single validation source existed for their cities [2], Toole et al. compare their estimates with travel surveys [1], and Bassolas et al. keep traffic counts out of the calibration but have them only at some main entrances of the city [41]. Second, the sources are usually tested together. The result then shows whether the complete model works, not which source made it work, even though uncertainty in demand affects every stage of a model [46, 47]. Third, a model needs a benchmark. Simple historical profiles predict the daily pattern of traffic very well [48, 49], so good timing means little unless a model does better than they do.

In this paper we build and test such a simulator for a whole region, Emilia-Romagna, Italy, with hourly traffic on the regional road network. We do not claim that it is more accurate than earlier ones, and no common benchmark would allow such a claim. Besides the simulator itself, we add a measure of the marginal information of each source: what changes when that source is added, changed or removed. We give each source one job in the model and check that job with data that played no part in it (Table 1). Half of the detectors scale the demand, with one factor per day type and one per province. Route choice uses GPS measured in one province and transferred to the rest of the region by road class. We compare timing on the GPS-matched roads of Forlì–Cesena, volumes at independent detectors in two seasons, and a local correction by leaving out each detector in turn, always against predictors that use no simulation. Section 2 describes the data and Section 3 the model and the tests. Section 4 follows the model as it is built: first the level of traffic, then its timing, the ranking of roads and the correction, then the error at single roads, and finally what each source adds. Section 5 discusses what this means.

Table 1 Role of each data source. Each step is compared with data that did not enter it; the depth of the correction is the one exception, explained in note *a*.

Data	Enters the model in	Used for comparison in
Mobile-phone OD (Jan–Feb 2020)	where and when trips occur	–
GPS traces, Forlì–Cesena	route choice (class speeds and usage)	–
GPS link counts, Forlì–Cesena, two campaigns	–	hourly profiles on the matched links
Detectors, half of February 2020	demand scale k, province factors, trip radius, departure shift, cost weights, depth[a]	–
Detectors, other half of February 2020	local correction; adoption of province factors[b]	volume level and ranking (2020)
Detectors, June–July 2021	local correction; adoption of province factors[b]	volume level and ranking, later season
Detector left out of the correction	–	leave-one-detector-out test

[a]The depth of the correction was selected by cross-validation over all detectors of 2020; 2021 is independent of it. [b]These detectors decided once whether the factors, set on the calibration half, were kept; they set no value.

Figure 1 shows how the data enter the simulator and which data are used to assess each output.

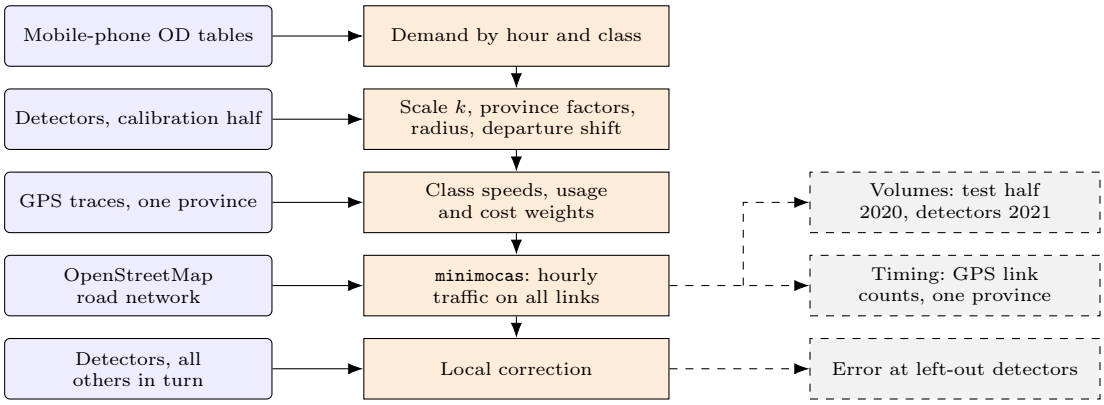


Fig. 1 Pipeline of the simulator. Blue: input data. Orange: steps of the model. Grey, dashed: data used to assess the output, which do not set the values of the steps they assess.

2 Data

The road network comes from OpenStreetMap [50, 51], whose road coverage has been estimated to exceed 95% in most European countries [52]. It has 3,925 links and 2,548 nodes, with their length, direction and turn restrictions, and each link has an OpenStreetMap road class (Fig. 2a). The network marks 122 links (151 km of 11,734 km) as closed, and simulated vehicles cannot use them. We kept these closures as they are, even though two detectors lie on closed links. One of the two, detector 153, records about 11,000 vehicles per day, so on that road either the closure or the position of the detector must be wrong.

For the demand, the operator TIM provided hourly activity between municipalities for 13 January–23 February 2020, which we averaged separately for each day of the week. The model keeps the relations among the 50 most active municipalities and between them and the provinces of neighbouring regions. When one end of a trip, or both, is outside this set, the model draws the missing end at random. We left out trips within each of the 50 municipalities, because adding them did not reduce the spread of the errors on the calibration detectors (Supplementary Section S2). Trips within the other municipalities stay in the demand, with both ends drawn at random.

So only part of the recorded activity enters the model as a relation. On a Monday, 20% of the activity between municipalities keeps its relation and about 50% has a recorded relation that the model does not use. The other 30% carries the destination code 000000000, which identifies no place, and we removed these cells from the table. Trips within a municipality are a further 44% of the table: 38% are within the 50 municipalities and are left out, and 5% are within other municipalities and enter with random ends. The result is 1.99 million agents on a Monday and 1.60 million on a Saturday. On a Monday, 27% have a known origin and destination and 73% have at least one random end (Fig. 3; Supplementary Section S1). The same averages are used for every simulated day, so eight simulations cover the fifteen days of the two campaigns, and the 2021 days use demand measured in early 2020. The hour labels in the tables refer to local time at the start of the hour (Supplementary Section S7).

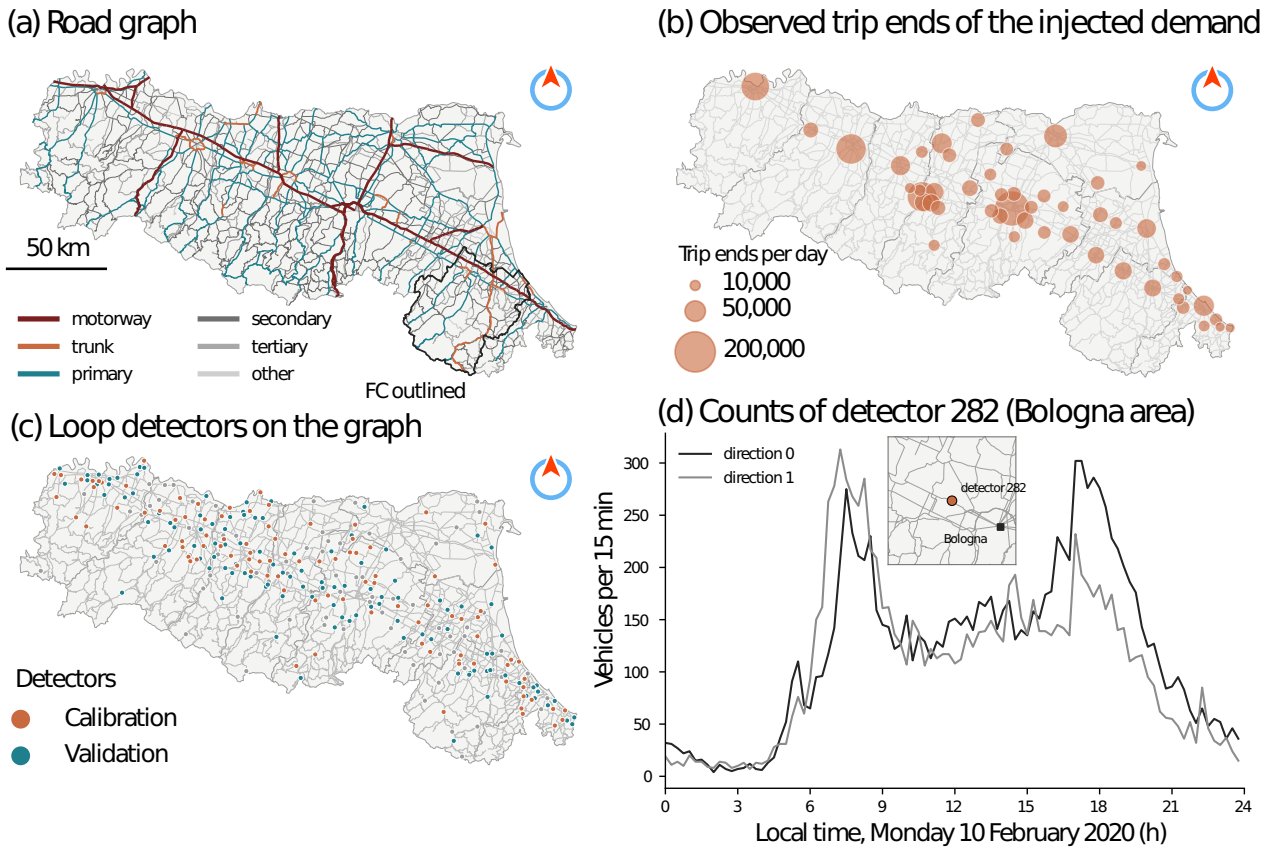


Fig. 2 The region and the data that enter the model. (a) Road network coloured by OpenStreetMap road class; the province of Forlì–Cesena, where GPS traces inform route choice, is outlined. **(b)** Daily trip ends with a known location in the weekday demand, by municipality: a larger circle means more trips, and the demand concentrates along the central axis of the region and on the coast. **(c)** Loop detectors of the regional MTS system placed on the network, by their use for daily volumes: the calibration half of February 2020 (100; 60 of them also enter the 2021 tests), validation only (115: the 97 of the test half and 18 used only in 2021), and others (77). **(d)** Counts of detector 282 near Bologna on 10 February 2020, 15-min resolution, local time. Road network © OpenStreetMap contributors.

The regional MTS system counts vehicles every 15 min in both directions, and 292 of its detectors lie on links of the network (Fig. 2c,d). For 2020 we use the project archive, and for 2021 the regional open-data release, which gives the same counts as the archive for February 2020. A detector is used on a given day only if

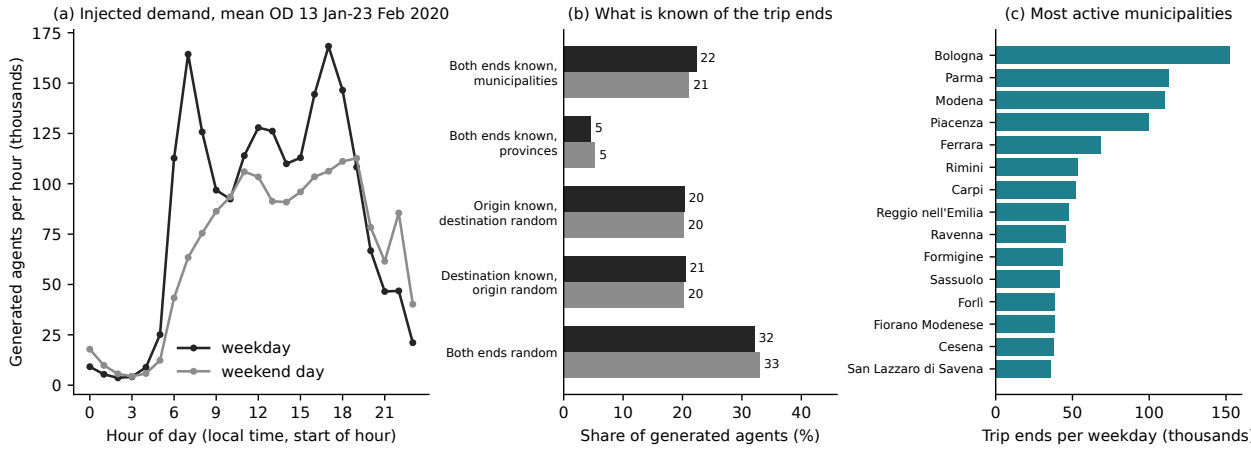


Fig. 3 Simulated demand. (a) Agents generated per hour for a weekday and a weekend day, after scaling by k and the province factors: weekdays show morning and evening peaks, and weekends a single broad rise through the day. **(b)** Shares of agents by what is known of their trip ends; "provinces" are relations with provinces of neighbouring regions. About a quarter of agents have both ends known, and the largest group has both ends drawn at random. **(c)** Municipalities with the most trip ends on a weekday.

both of its directions are complete. This filter favours busy roads, but relaxing it hardly changes the errors (Supplementary Section S7).

Viasat provided hourly probe counts per matched link for two campaigns, 10–17 February 2020 (six weekdays and a weekend) and 28 June–4 July 2021 (five weekdays and a weekend), and the raw traces of private vehicles in the province of Forlì–Cesena. We use the link counts of that province only (Supplementary Section S5). Probes were not used to generate or scale the demand, or in the detector correction.

Phone data were provided as anonymized counts aggregated in space and time, and we accessed no individual records. GPS data were provided as pseudonymized vehicle records and used only to compute link counts, speeds and road-class aggregates. No personal data are reported, and ethics approval was not required.

3 Methods

3.1 Simulation and route choice

The simulator moves agents in steps of 10 s, each at the speed of its link. This speed falls as the link becomes denser, down to a critical speed. Agents with a known origin and destination follow best paths computed with Dijkstra's algorithm [53], which respect turn restrictions. Agents with a random end instead choose the next link at each node, with a probability proportional to a transition weight, and travel for at most 5,600 s. For each hour and direction the simulator records how many distinct agents used each link, while virtual counters at the detector locations record crossings. Each run starts empty at midnight, and a given random seed always gives the same run. Traffic is not assigned to an equilibrium [26], and the model does not reproduce the congestion phenomena of microscopic traffic-flow models [54–56].

To inform route choice, we matched GPS fixes of private vehicles in Forlì–Cesena (February 2020) to the road network. For each road class we took the median off-peak speed and the median daily number of probes per link (Supplementary Section S3), and applied both to all links of that class in the region. The speeds seem stable over time: on the same roads, those of June 2021 have a rank correlation of 0.98 with those of February 2020. The probe numbers set the transition weights of agents with a random end. They also enter the best-path cost of link i,

$$w_i = L_i \left[\alpha + \alpha_s \frac{v_{\min}}{v_i} + (1 - \alpha - \alpha_s) \frac{1/c_i - 1/c_{\max}}{1/c_{\min} - 1/c_{\max}} \right], \tag{1}$$

where L_i is its length, v_i its speed, $v_{\min} = 20$ km/h, and c_i the usage weight of its class, so that busier road classes cost less. This cost describes an empirical preference by road class, and it is not a behavioural model. We chose the usage-weighted form among shortest-distance, travel-time and mixed alternatives because of its rank correlation between simulated daily volumes and GPS counts in Forlì–Cesena, without using detectors. The weights $\alpha = 0.5$ and $\alpha_s = 0.25$ come from a discrete grid compared on the calibration detectors; they are chosen values, not estimates (Supplementary Section S3).

3.2 Demand scaling and local correction

The OD tables describe one operator's customers, not all traffic, so the number of agents has to be scaled. We multiply it by a factor k. This factor combines coverage, vehicle occupancy and the assumptions on random trip ends, and has no physical meaning of its own. To set it, we split the detectors with complete data in February 2020 into a calibration half (100 detectors) and a test half (97), with the same share of each province in both.

Weekday counts are averaged and compared with a Monday simulation, and weekend counts with a Saturday simulation. For each day type, we choose the k that minimizes the mean squared $\log_{10}$(simulated/observed) of daily volumes, on a grid from 0.9 to 1.8 (Supplementary Section S2). The test half and the 2021 campaign are not used in this choice.

In this paper, *dispersion* means the standard deviation of $\log_{10}$(simulated/observed) of daily volumes across detectors. It measures how much the error changes from road to road once the overall level is set.

Origins are drawn around the source point, random ends lie within 25 km of the known end, and departures are 45 min earlier than the OD label. The radius was chosen together with k, by the same loss, among 15, 20, 25 and 30 km. The shift is an empirical adjustment on the calibration detectors, confirmed on weekdays (Supplementary Section S2).

Dispersion on the calibration half was still large after these steps, and 37% of its variance was between provinces (Section 4.3), so we tested one more factor per province. Each trip source with a known location is multiplied by the geometric mean of the factors of its ends inside the region. Sources without a location and ends outside the region keep a factor of one, and k does not change. The factor of a province starts from one and is multiplied by 10^{-m}, where m is the median $\log_{10}$(simulated/observed) of its calibration detectors over both day types. We do this for two rounds. We fixed the rule for adopting the factors before the runs: the 95% paired bootstrap interval of the change in dispersion had to be below zero on the test half and, only if it was, on the detectors of 2021. The factors add nine parameters, and their values come from the calibration half only.

Detectors also correct the simulation locally. Where there is a detector, we can replace the simulated count with the observed one and spread the difference to nearby roads. Let $u_i(h)$ be the simulated count on the directed link of detector i in hour h and $c_i(h)$ the observed count in that direction; detectors on the same directed link are averaged. The increment at the detector link is

$$\delta_i(h) = \begin{cases} c_i(h) - u_i(h) & \text{if } u_i(h) \geq 1, \\ 0 & \text{otherwise,} \end{cases} \tag{2}$$

so the corrected count $\widetilde{u}_i(h) = u_i(h) + \delta_i(h)$ equals the observed one wherever the simulation has traffic at the detector. From there the increment spreads downstream and upstream. Each adjacent link receives the increment of the previous link, multiplied by its own simulated share of the flows leaving (downstream) or entering (upstream) the common node, $\delta_{(n_2,n)} = \delta_{(n_1,n_2)}\, u_{(n_2,n)} / \sum_m u_{(n_2,m)}$. The spreading does not enter other detector links. It stops when the increment falls below one vehicle, after three links along a path, at a forbidden turn or at a closed link. Any negative count is set to zero.

3.3 How the model is assessed

To assess timing, for each GPS-matched link we compare the hourly shares $p_i(h)$ and $q_i(h)$ of the reconstructed and GPS counts over the campaign days, with both directions together, using

$$\mathrm{MAE}_i = \frac{100}{24} \sum_{h=0}^{23} |p_i(h) - q_i(h)|, \tag{3}$$

in percentage points (pp). We use three benchmarks, each applied to every link: the OD profile of the demand, the mean profile of the detectors and a flat profile.

We compare daily volumes in both directions at detectors with complete data, using the ratio of total simulated to total observed volume, the Spearman rank correlation, the median absolute $\log_{10}$ ratio and the share of detectors within a factor of two. The 2020 volume of a detector is its mean over the weekdays or the weekend days of February 2020. Intervals come from a paired bootstrap over detectors.

The predictors without simulation take the level from the median volume of other detectors of the same road class and the profile from the mean profile of other detectors. For the test half, both are built on the calibration half. For the leave-one-out comparison they are built, like the correction itself, on the other detectors of the same day.

To test the local correction, we leave out all detectors of a link in turn and compare their counts with the corrected reconstruction on that link. The error is $\sum |\widetilde{u}(h) - c(h)| / \sum c(h)$ over hours, directions and days. The correction also has a depth, the number of links it spreads along. We use the smallest depth whose error reduction, in 5-fold cross-validation over the detectors of 2020, is within one bootstrap standard error of the best depth. Further details are in Supplementary Section S7.

Experiments on the OD relations, which separate them from the province factors and rearrange trips with the numbers by place preserved, are described with their results in Supplementary Section S9. They followed protocols written before their runs, are exploratory, because the data had already been seen, and do not change the model.

The last tests use only data not used to set the model: the raw simulation on the test half of 2020 and on all detectors of 2021. Each test looks at one property, and passing it supports only that property. T1 is about the level: the volume ratio counts as equivalent to one within 0.80–1.25 if the 90% bootstrap intervals, over detectors and over provinces, are both within that margin, as in two one-sided tests. Since we chose this margin ourselves, we also report the smallest one that passes. T2 and T3 are about ranking. T2 asks whether the Spearman correlation is higher than chance, by permuting simulated volumes among detectors. T3 permutes them only within road class, so it looks only at what road class alone does not explain. T4 is about timing: on the links of Forlì–Cesena, the hourly-share error should be below that of the flat profile (one-sided Wilcoxon signed-rank test). P values are Holm-adjusted over the six tests. The data have three limits as independent evidence. The test half and 2021 decided whether to adopt the province factors. The 2021 set includes 60 detectors of the calibration half, on the same roads, so we also report it without them. Finally, these data were not blind: the removal of the destination code and the use of a travel-time term in the cost were decided after earlier versions of the model had been compared on them. Numerical values were always set on the calibration half, as Table 1 shows, but the results are held out rather than pre-registered.

4 Results

4.1 Level and timing of traffic

We first ask whether the simulator puts the right amount of traffic on the network. Scaling the demand by day type, with $k = 1.4$ on weekdays and $k = 1.2$ on weekends, brings simulated and observed volume on the calibration detectors within 4% and 7%, and within 1% and 4% with the province factors. A single factor of 1.3 would leave weekdays 3% too low and weekends 16% too high (Supplementary Fig. S1). The selected values remain the minima of the grid with province factors, and subsets of 50 calibration detectors choose the same value or a neighbour in nearly all cases (Supplementary Section S8). Regional differences remain after this step, and the province factors correct them. Figure 4a shows that the correction is far from uniform: demand is raised in the north-west, in Modena and in Ferrara and lowered along the east and in Reggio Emilia. We treat these factors as empirical, not as measured differences in phone coverage or car use.

Figures 4b and 4c show the outcome at detectors that did not set these values, or set them only in part in 2021. In both campaigns the cloud of points follows the diagonal: the overall level is right and busy roads tend to be simulated as busy, while the width of the cloud shows the error at single roads. Most roads lie within a factor of two of their count. Table 2 shows what the province factors contribute. Their gain is clearest at the typical detector, where the median error falls by about a fifth, and the rule for adopting them was met in both data sets. Factors estimated on 25 or 50 calibration detectors mostly give a dispersion close to that of the 100 used here, although the value for a single province varies strongly between subsets, so the factors should be read as effective corrections and not as precise estimates (Supplementary Section S8). All later results use the model with the factors.

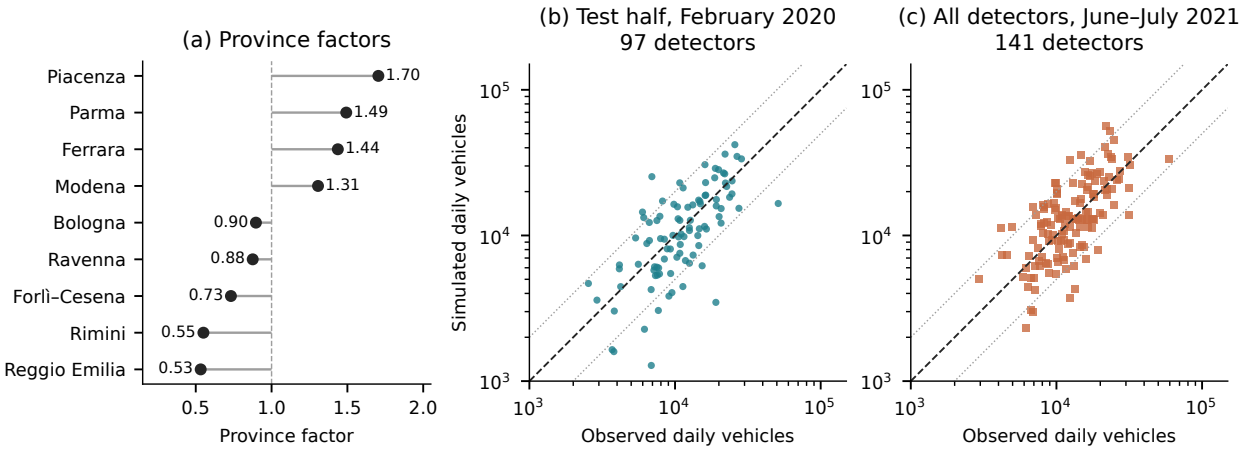


Fig. 4 The simulator reaches the overall level of traffic. (a) Factor applied to the demand of each province; values above one (dashed line) raise the demand. The factors were set on the calibration half of the detectors. **(b,c)** Daily volume at each detector, simulated against observed, raw simulation before any local correction, on logarithmic axes. A point on the dashed line would be a perfect estimate, and points between the dotted lines are within a factor of two. A cloud centred on the dashed line means that the level is right overall, and its width shows the error at single roads. Panel (b) uses detectors kept out of all calibration; panel (c) uses a later season and includes 60 calibration detectors. One detector in each panel has no simulated traffic and is not drawn.

With the level set, we turn to when traffic occurs, using GPS as a view of the daily rhythm on the matched roads of Forlì–Cesena. Figure 5a,b compares the simulation with two simple alternatives: the hourly profile of the demand itself and the average profile of the detectors. All three are far better than a flat profile, and the

Table 2 Effect of the province factors, without and with them. Dispersion: standard deviation of $\log_{10}$(simulated/observed) of daily volumes across detectors; the test-half values include detector 153, which has no simulated traffic. Median error: median absolute $\log_{10}$ ratio per detector and day type. Intervals: 95% paired bootstrap of the change.

	Calibration half		Test half, 2020		2021	
Measure	without	with	without	with	without	with
Dispersion	0.217	0.184	0.498	0.488	0.412	0.399
change, interval	$[-0.047, -0.018]$		$[-0.033, -0.002]$		$[-0.034, -0.006]$	
Spearman correlation	0.68	0.74	0.63	0.67	0.63	0.66
change, interval	$[0.02, 0.12]$		$[-0.01, 0.08]$		$[-0.02, 0.07]$	
Median error	0.151	0.100	0.179	0.143	0.178	0.143
Volume ratio	1.05	1.03	1.04	1.01	1.15	1.10

curves are close: most of the daily rhythm is already in the demand. The simulation improves on the demand profile, and after correction it matches the detector average in 2020 and stays slightly behind it in 2021 (Table 3). Weekend timing is harder than weekday timing for both the simulation and the detector average (Supplementary Fig. S4). GPS traces of this province also set the route weights, so this comparison is not fully independent of route choice, although route choice hardly changes the profiles (Supplementary Section S3). Nearby links are correlated, so the intervals behind these comparisons are optimistic (Supplementary Section S5).

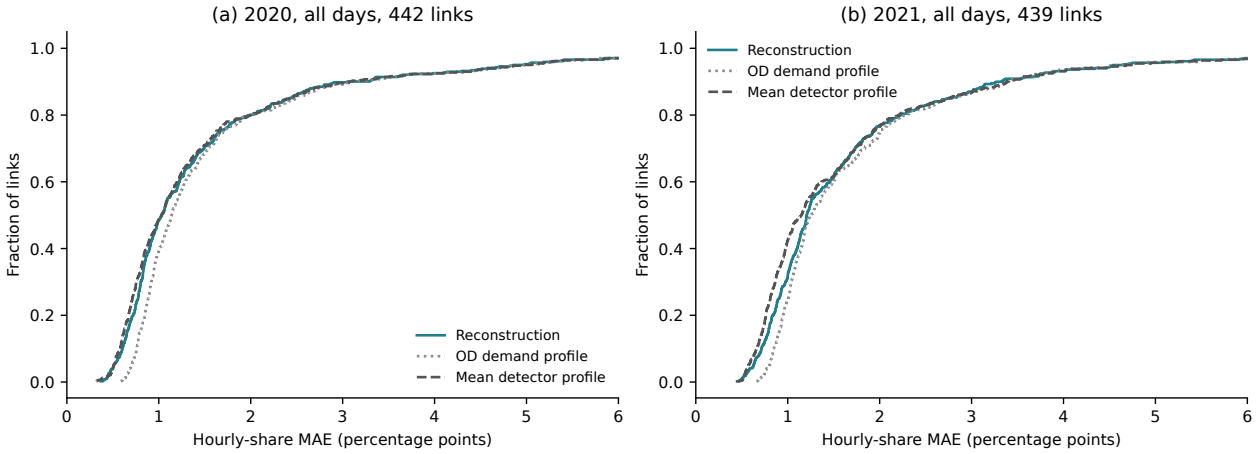


Fig. 5 The timing of traffic follows GPS on nearly all links. For each GPS-matched link we compute the error of its hourly shares, the mean absolute difference between the share of daily traffic in each hour in the estimate and in GPS. Each curve shows the fraction of links whose error is below the value on the horizontal axis, so a curve further up and to the left is better. Reconstruction: simulation after local correction. **(a,b)** All days, against the OD demand profile and the mean detector profile, in 2020 and 2021. The curves are close, and a minority of links has large errors for all three. Curves continue beyond the 6 pp limit of the axis.

Table 3 Median hourly-share error (pp) on the GPS links of Forlì–Cesena, all days of each campaign; lower is better.

Campaign	Links	Raw	Corrected	Mean detector	OD	Flat
2020	442	1.06	1.03	1.03	1.14	2.72
2021	439	1.21	1.21	1.15	1.25	2.60

Detectors count all vehicles, so they let us look at the actual flow through the day. Figure 6 shows the simulator at its best, at the four test detectors with the lowest error. On weekdays it follows both peaks and the midday plateau. On weekends it follows the general shape, starting too low at night and running too high late in the evening. These are selected cases, and Table 4 gives the picture over all test detectors. The hourly error is about 40%, and it would fall by about two thirds on weekdays if each detector were given its observed daily level, so at a single road the difference is mostly how much traffic there is, not when. All detectors share a similar daily pattern, and the plain average of the calibration detectors has a higher profile correlation with each test detector than the simulation has. The simulation, however, ranks the roads clearly better than a predictor based on road class.

4.2 Daily volumes and local correction

Since most of the difference at detectors is about level, we now look at daily volumes (Table 5, top). The results of the two campaigns are similar: the total volume is close to the observed one, the ranking of roads is

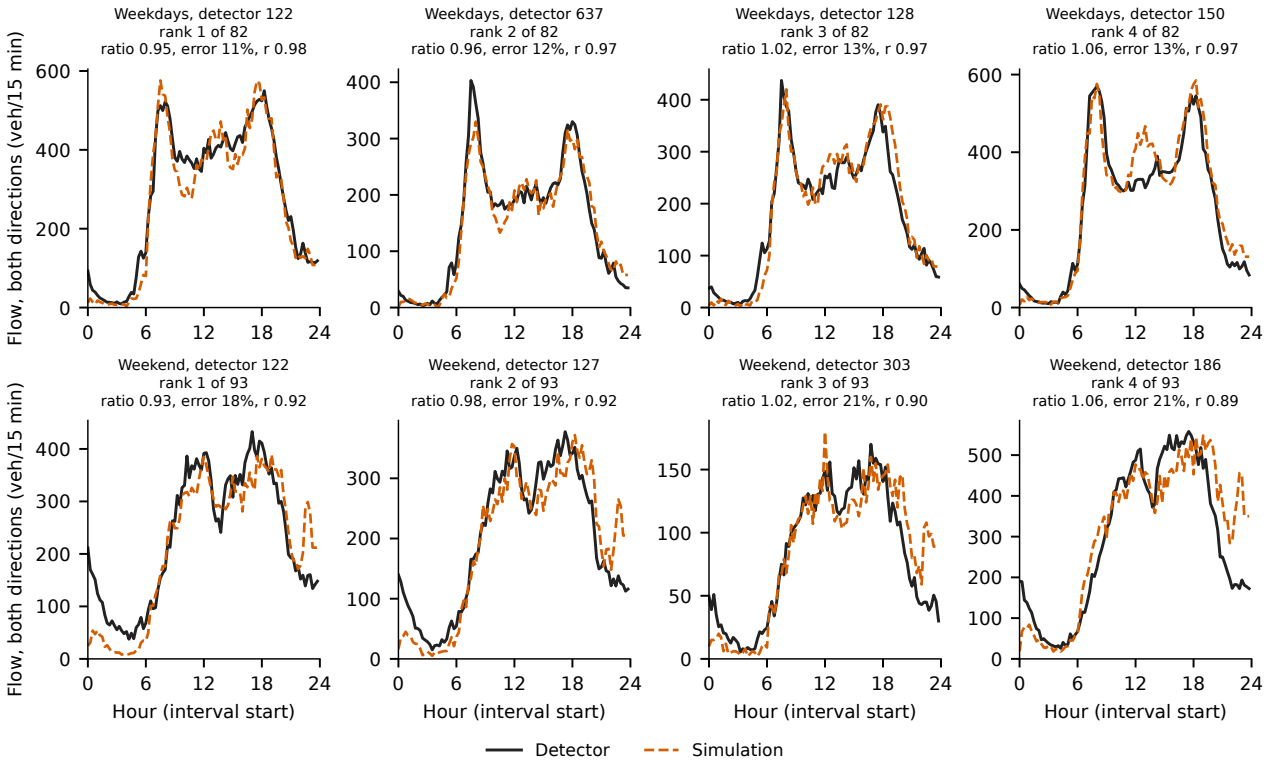


Fig. 6 Where the simulator works best: the four test detectors with the lowest error. Mean observed flow (solid) and raw simulation before any correction (dashed), both directions, February 2020, 15-min resolution, for weekdays (top) and weekends (bottom). The title of each panel gives the rank among test detectors, the ratio of simulated to observed volume, the 15-min error and the correlation of the two curves. These are the best cases, selected for their low error: the median detector has a 15-min error of 33% on weekdays and 38% on weekends.

Table 4 Raw simulation against a predictor without simulation, test detectors of February 2020. The predictor, built on the calibration detectors, is the median volume of the same road class times the mean profile. The province factors were also set on those detectors. Volumes and errors come from the virtual counters, both directions; these counters are about 8% below the distinct-agent counts on which k is set. Intervals: 95% bootstrap of the difference, simulation minus predictor. Peaks, simulated against observed: weekdays 8:00 and 18:00 against 7:30 and 17:30; weekend evening 19:15 against 17:30.

	Weekdays (82 detectors)			Weekends (93 detectors)		
Measure	Sim.	Pred.	Interval	Sim.	Pred.	Interval
Volume ratio	0.93	0.92		0.93	0.88	
Median $\lvert \log_{10} \rvert$ ratio, daily volume	0.152	0.137	$[-0.05, +0.05]$	0.147	0.157	$[-0.07, +0.02]$
Within a factor of two (%)	83	89		83	88	
Spearman, daily volume	0.67	0.41	$[0.06, 0.46]$	0.68	0.48	$[0.02, 0.36]$
Hourly error (%)	37.6	40.1	$[-11.5, +6.0]$	42.8	41.1	$[-6.2, +9.4]$
Hourly error with known level (%)	12.8	7.2		22.1	8.2	
Median profile correlation	0.96	0.99	$[-0.03, -0.02]$	0.87	0.98	$[-0.12, -0.09]$

good and more than four detectors in five are within a factor of two. The level in 2021 is reached with demand measured in early 2020 and a scale fixed on winter counts, so it should not be read as a test of demand in that season. Without the 60 calibration detectors, the 81 remaining detectors of 2021 give a volume ratio of 1.11 [0.99, 1.23], a Spearman correlation of 0.65 [0.50, 0.76] and 83% [74, 90] within a factor of two. Road class adds a further pattern (Fig. 7): the simulation tends to underestimate trunk roads and to overestimate primary roads slightly, although few detectors lie on trunk roads.

The contribution of GPS appears when route choice is changed under the same demand (Table 5, bottom, and Fig. 9a). At detectors outside the province where route choice was set, GPS-informed routing ranks roads better than shortest-distance routing in both campaigns, while the typical error does not change measurably. This gain comes mainly from the use of each road class, and GPS speeds alone add little (Supplementary Section S8). Repeating the selection on the final model confirms the chosen form. A simple predictor gives a useful comparison: road class and the mean profile, with no simulation, give a similar error on daily volumes, and the simulation does better in ranking the detectors (Table 4).

All results so far describe the raw simulation. We now add the local correction. To test it fairly, we leave out each detector in turn and ask the correction, which uses the other detectors, to predict its count. The correction removes 2.5 percentage points of the error in 2020 and 6.2 in 2021 at detectors it has not seen (Table 6), and

Table 5 Daily volumes at independent detectors, raw simulation, with 95% bootstrap intervals. Top: all detectors of each set. Bottom: GPS-informed against shortest-distance routing under the same demand, at test detectors outside Forlì–Cesena; intervals are for the paired difference.

Measure	Test half, 2020 (97 detectors)	2021 (141 detectors)
Volume ratio	1.00 [0.89, 1.11]	1.10 [1.01, 1.19]
Spearman correlation	0.70 [0.56, 0.80]	0.66 [0.54, 0.75]
Median $\lvert \log_{10} \rvert$ ratio	0.151 [0.112, 0.173]	0.143 [0.119, 0.176]
Within a factor of two (%)	81 [73, 90]	85 [79, 91]
	Test half, outside FC (87)	2021, outside FC (128)
Spearman, shortest distance to GPS	0.63 to 0.72, +0.09 [0.02, 0.17]	0.53 to 0.64, +0.10 [0.03, 0.18]
Median $\lvert \log_{10} \rvert$ ratio, shortest distance to GPS	0.147 to 0.141, −0.01 [−0.05, 0.03]	0.152 to 0.142, −0.01 [−0.04, 0.04]

the spread between detectors remains larger than the gain. About three detectors in five improve and about one in four gets worse. The 2020 set includes the calibration detectors and the depth of the correction was selected on 2020, so 2021 is the independent estimate (Supplementary Section S4). With a fixed 20% held out on single days, the gain grows with the number of detectors, from 1.5 to 3.3 pp on the winter day and from 0.0 to 1.5 pp on the summer day (Supplementary Section S8).

Again we compare with the predictor built on the other detectors of the same day (Table 6 and Fig. 9c). The corrected reconstruction cannot be told apart from it, and each is better at about half of the detectors. The simulation, in addition, gives a complete and consistent field over all links of the network.

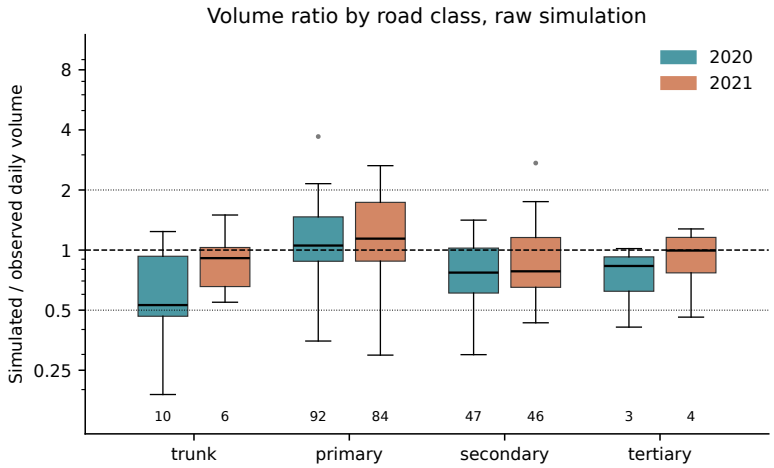


Fig. 7 Simulated volume by road class. Ratio of simulated to observed daily volume, raw simulation, with the number of detectors below each box. The dashed line marks perfect agreement and the dotted lines a factor of two; a box below the dashed line means that the class is underestimated. Trunk and tertiary roads have few detectors. One detector with zero simulated traffic is not shown. Boxes: quartiles and median; whiskers: 1.5 interquartile ranges.

Table 6 Local correction at left-out detectors, depth 3. Hourly error pooled over hours, directions and days. Predictor: road class and mean profile from the other detectors of the same day. Intervals: 95% bootstrap.

Measure	2020 (152 detectors)	2021 (141 detectors)
Hourly error, raw (%)	42.2	47.6
Hourly error, corrected (%)	39.7	41.4
Hourly error, predictor (%)	36.9	37.6
Reduction by the correction (pp)	2.5 [0.3, 5.0]	6.2 [2.6, 10.6]
Raw minus predictor (pp)	[−1.3, +11.7]	[+1.9, +19.1]
Corrected minus predictor (pp)	[−3.1, +8.7]	[−3.0, +10.9]
Median detector, raw to corrected (%)	35.8 to 33.2	38.4 to 35.5
Detectors improved / worsened (%)	60 / 26	63 / 22
Detectors better than the predictor (%)	46	50
Spearman of volumes, raw to corrected	0.72 to 0.73	0.66 to 0.69

With the correction we can draw the full traffic field (Fig. 8b). The motorway axis and the roads around the main cities carry most traffic, and the Apennine roads in the south carry little. Figure 8a shows the best case of the test: with detector 349 left out, the reconstruction on its road follows the daily curve and misses the daily total by about 2%. This is what the simulator can achieve where it works best (Table 6 gives all detectors). Figure 9 collects the gains and differences discussed so far, together with the OD relations.

We tested the OD relations in detail in Supplementary Section S9. Removing them lowers the ranking of roads on the test half, but this also removes the regional pattern of the province factors. With that pattern kept, no gain in ranking can be established, and the spread of errors is larger with the relations. Destinations rearranged at random make trips far too long and tend to raise the spread, while destinations rearranged at the same distances cannot be told apart from the real ones. These tests have limited power, because link counts respond little to pairings.

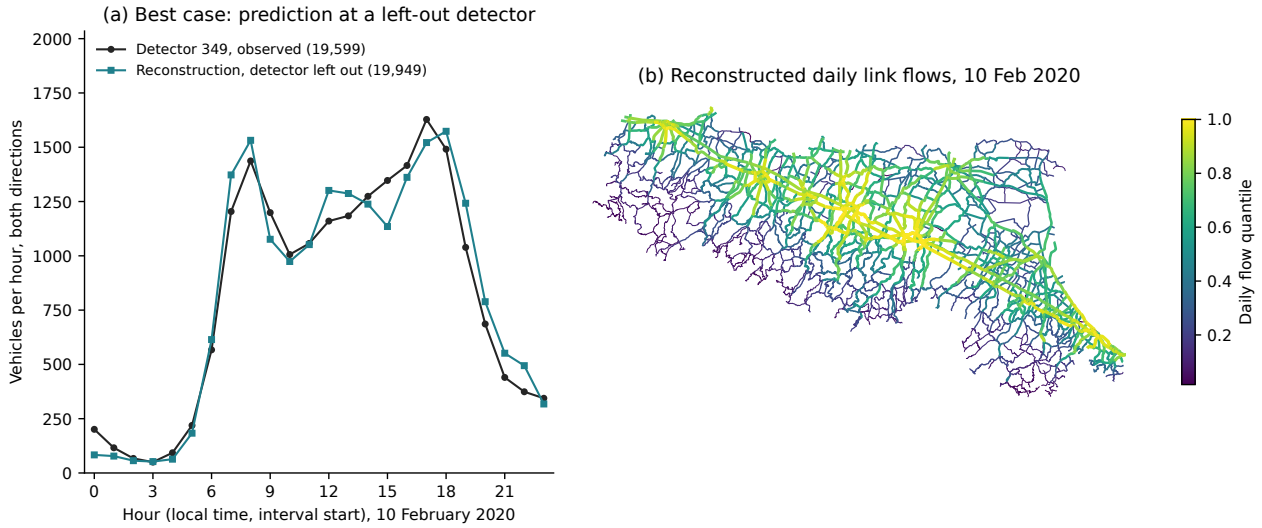


Fig. 8 Reconstructed traffic field. (a) The best case among the 61 detectors complete on 10 February 2020: hourly counts of detector 349 (both directions) and the reconstruction on its link when that detector is left out of the correction and all others are used; daily totals in parentheses. The error is 10%, against a median of 34% over the 61 detectors. **(b)** Reconstructed daily flow on each link, with all detectors used. Colour and line width show the quantile of the flow, so yellow marks the busiest roads of the region.

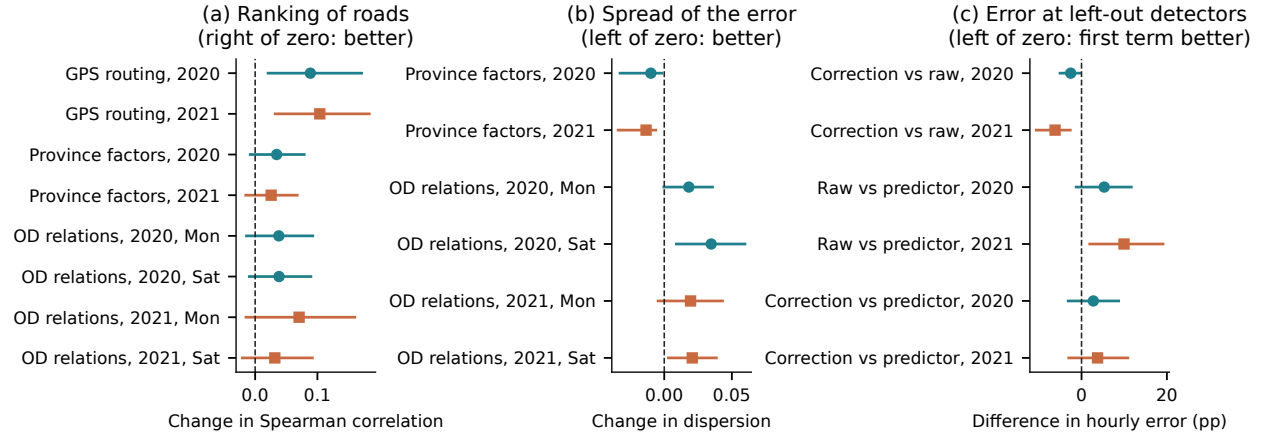


Fig. 9 The marginal information of each source. Each row is one comparison on data not used to set the values: the marker is the estimated change and the bar its 95% bootstrap interval. A bar that crosses the dashed zero line means that the data cannot establish a change. Blue circles: 2020; orange squares: 2021. For the OD relations, the regional pattern of the factors is kept in the simulation without relations (Supplementary Section S9). **(a)** Change in the rank correlation between simulated and observed volumes when the source is added. GPS routing helps in both campaigns; no interval of the OD relations clears zero. **(b)** Change in dispersion, the spread of the error between roads. The province factors reduce it, and the OD relations do not. **(c)** Differences in hourly error at left-out detectors. The correction beats the raw simulation, but not the predictor without simulation.

4.3 Error at single roads

The results so far leave a large difference at single roads, so we look at what kind of error it is. Figure 10a,b shows that it is not noise. A road that the simulator overestimates on weekdays is overestimated by about the same amount on weekends ($r = 0.95$), and the same holds between winter 2020 and summer 2021 ($r = 0.98$). The day-to-day variation of the counts is about ten times smaller than this error. Part of the stability between seasons is expected, since both campaigns use the same demand; it also means that the counts on each road changed little relative to the simulated ones.

Figures 10c and 10d show what the province factors did to this error. Without them, province was its largest known component and nearby detectors had similar errors. With them, that regional component has mostly gone and nearby errors are no more alike than chance would give. Together, province, road class and

volume now explain about a quarter of the variance. Misplaced detectors do not explain the rest: comparing their coordinates with the road geometry, we flagged 5 of 292 as doubtful, and dispersion remains large without them (Supplementary Section S5). A further analysis, out of sample, shows that road class is the only group of road properties with a repeatable signal, whereas nearby population and activities, centrality, parallel roads and the density of OpenStreetMap roads not in the network do not explain the error in a way that carries over between provinces (Supplementary Section S8). Route weights from the GPS count of each link, instead of its class, did not give a clear gain at the detectors of Forlì–Cesena, on few detectors (Supplementary Section S8). What remains is a stable property of each road, which may reflect how demand reaches the road locally, and which these data cannot separate further.

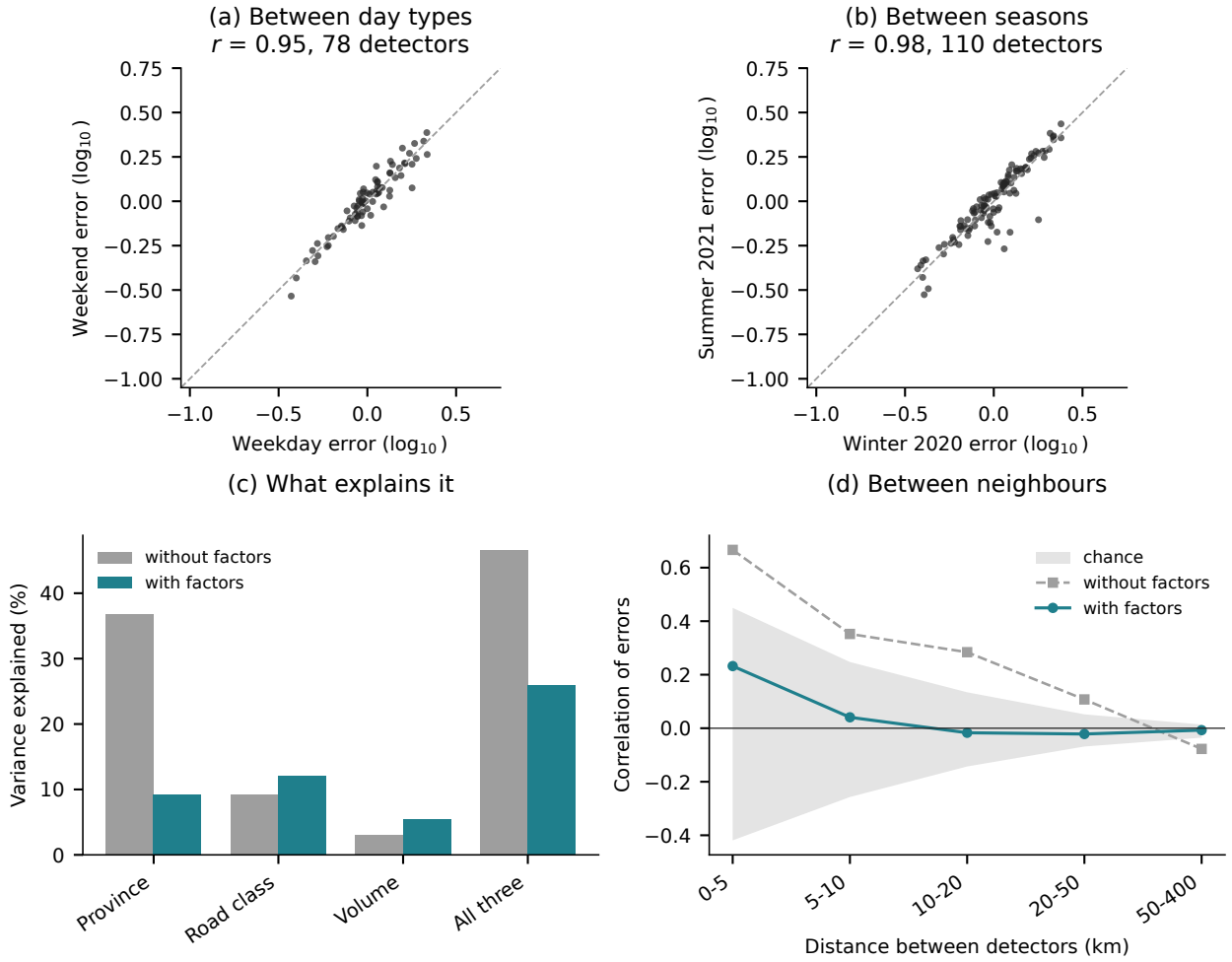


Fig. 10 The error at a single road is stable and specific to that road. The error is $\log_{10}$(simulated/observed) of daily volume, so zero is perfect and 0.3 is about a factor of two. **(a)** Error of each calibration detector on weekdays against weekends. **(b)** Error of each detector in winter 2020 against summer 2021. Points close to the dashed diagonal have the same error in both conditions. **(c)** Share of the variance of the error across calibration detectors explained by province, road class, observed volume and all three together, on the Monday, without and with province factors. **(d)** Correlation between the errors of pairs of detectors, by distance. The shaded band is the 95% range expected by chance, from permutations with the factors; a point inside it shows no detectable similarity between neighbours.

4.4 Formal tests and contribution of each source

Finally, we test each property on independent data (Table 7). The overall level is equivalent to the observed one within the chosen margin, the ranking of roads exceeds chance even when road class is held fixed, and the timing beats a flat profile on nearly all links of Forlì–Cesena. A regression across detectors gives a slope close to one, with much of the variance left to the single-road error described above. The province interval of T1 is based on few provinces, each of which received its own factor, so it is only partly independent.

Table 8 collects the effect of each source on data not used to set it. Phone data give the timing and, with detector scaling, the level; GPS improves the ranking of roads; detectors also bring a local gain through the correction. The OD tables add the short-range structure of trips, and we could not show a further value for specific pairings.

Table 7 Final tests on data not used to set the model values, raw simulation. Intervals for T1 are 90%; for the slope, 95%. FC: Forlì–Cesena. The smallest margin of T1 that passes in both campaigns is about 1.21. Holm-adjusted $p < 0.001$ for T2 and T3. In T4 the simulation beats the flat profile on more than 99% of links. The 2021 set includes 60 calibration detectors.

Test	Test half, 2020 (97 detectors)	2021 (141 detectors)
T1 Volume ratio, resampling detectors	1.00 [0.91, 1.09]: pass	1.10 [1.02, 1.18]: pass
T1 Volume ratio, resampling provinces	[0.86, 1.13]: pass	[0.97, 1.21]: pass
T2 Spearman above chance	0.70, $p < 0.001$	0.66, $p < 0.001$
T3 Spearman within road class	$p < 0.001$	$p < 0.001$
T4 Timing against flat, FC links (error, pp)	1.06 against 2.72, $p < 10^{-70}$	1.21 against 2.60, $p < 10^{-70}$
Slope, log–log volumes (R^2)	0.94 [0.72, 1.18] (0.19)	0.86 [0.70, 1.05] (0.17)

Table 8 What each source adds, on data not used to set it. Differences are in the sense of adding the source; negative errors are improvements. Intervals are 95% bootstrap.

Source	Role	Effect
Mobile-phone OD, total and timing	agents per interval	hourly-share error on FC links 1.06 and 1.21 pp against 2.72 and 2.60 flat; 0.06 and 0.07 pp lower than the OD profile; mean detector profile 1.03 and 1.15 pp
Mobile-phone OD, relations as used (20% of activity between municipalities)	known trip ends	with the factors kept on both sides: Spearman +0.03 to +0.07, intervals include zero; dispersion +0.018 to +0.035 (worse), two of four intervals exclude zero. Against random pairs: dispersion −0.016 to −0.038, two of four intervals exclude zero. Against pairs with the same distances: one of four intervals excludes zero at 50 municipalities, none at 347
Detectors, half	scale k by day type	test ratio 1.00; 1.10 in 2021
Detectors, half (nine values)	province factors	median $\lvert \log_{10} \rvert$ ratio per detector and day type 0.179 to 0.143 on the test half, 0.178 to 0.143 in 2021; dispersion −0.010 and −0.013 (intervals exclude zero)
GPS traces	route weights	Spearman +0.09 [0.02, 0.17] and +0.10 [0.03, 0.18] outside the province; median error not measurably different; profiles unchanged
Detectors, all	local correction	hourly error −2.5 pp [−5.0, −0.3] and −6.2 pp [−10.6, −2.6] at left-out detectors; profile error −0.01 and 0.00 pp on GPS links

5 Discussion

We built a simulator of ordinary regional traffic from three data sources. With eleven demand-level parameters (two by day type, nine by province), it reaches the observed overall level and timing, it ranks roads well above chance, and each source has a measurable role. Detectors set the level and reduce the regional bias. GPS use by road class improves the ranking of roads. Phone tables give the daily timing and the short-range structure of trips. The correction brings a further local gain. The province factors are empirical: a factor of 1.70 or 0.53 cannot be read as a difference in phone coverage or car use without further evidence, and the factors would have to be estimated again for another period or region.

The OD tables need careful reading, because our tests answer narrower questions than "do OD data have value?". Destinations rearranged at random make trips far too long and tend to worsen the model, so the tables carry useful information on trip length. Random ends drawn near the known end supply much of the same information, which may be why removing the relations does not raise dispersion; this explanation is a hypothesis. Beyond distance, we detect no value in specific pairings, in a setting where about one fifth of inter-municipal trips keep their relation, those relations are shorter than the 25-km radius around their ends, and link counts respond little to pairings. The OD pairings may therefore be useful where we could not test them, for example in applications that need trips rather than link volumes. In the same way, a value of $k = 1.4$ cannot be read as the inverse of the operator's coverage.

At the detectors, the simulation has an error similar to that of a predictor without simulation, and it ranks the test half better. That predictor gives no routes and no traffic on roads without detectors, which the simulator does provide. Detectors are not a random sample of roads.

Three findings may apply beyond this region, as hypotheses for other settings. Timing needs a benchmark, since a common daily profile already explains most of the hourly pattern. A few scale parameters can reach the overall level, so the error at single roads should be reported along with it. Removing a source can mislead when other parameters act through it, so its effect should be tested with those parameters kept, and with the

marginals preserved. The values we report are marginal to this model and to the order in which we built it, and they are not an exchangeable decomposition. They also concern ordinary traffic.

Some boundaries of these results should be kept in mind. GPS comparison covers one province, which also informs route choice. The experiments on the OD relations are exploratory and rest on one simulated Monday and Saturday per campaign. The test half and 2021 took part in the decision on the province factors and had been seen when the final configuration was fixed. Several settings (trip radius, cost weights, origin spreading) were chosen on the calibration half without a demonstrated gain on independent detectors. The largest remaining error is the level at individual roads, so aggregate agreement should not be read as accuracy at a given road.

This paper establishes what the simulator does for ordinary traffic: regional level, ranking of roads and timing, and hourly volumes on roads without detectors that are consistent with the network. It can therefore serve as a computational basis for later work on road closures, demand perturbations and network resilience. We make no claim here about those applications: skill on ordinary traffic does not imply skill under a change, and they need their own assessment on observed changes.

6 Conclusions

We have built a regional traffic simulator from mobile-phone demand, detector counts and GPS, and measured what each source adds on data not used to set each step. Scaling by day type and province gives the observed overall volume on independent detectors and reduces the regional bias. Route weights based on GPS use improve the ranking of roads, and the local correction reduces the error at left-out detectors. The overall timing comes from the daily profile of the demand. The OD tables contribute the level, the timing and the short-range structure of trips; on these link counts, specific pairings add no detectable value beyond that structure, and the test has limited power. Tests on independent data support the overall level, the ranking of roads and the timing. At single roads the error is stable and specific to each road, and the simulation is comparable to predictors based on road class and other detectors. A simulator of this type gives a consistent picture of ordinary traffic at the scale of the region, over all links of the network, and its use for scenarios remains future work that needs its own assessment on observed changes.

Supplementary information

Additional file 1 describes the data layers and the demand classes (S1), the demand scale and the changes tested on the calibration detectors (S2), the simulation model and the route-choice variants (S3), the detector correction and its sensitivity to depth (S4), additional results (S5), the final configuration and the reproduction of all results (S6), further details of data and methods (S7) the experiments on the final model (S8) and the value of the OD relations (S9).

Abbreviations

GPS, Global Positioning System; MAE, mean absolute error; MTS, regional traffic monitoring system of Emilia-Romagna; OD, origin–destination; pp, percentage points.

Declarations

Availability of data and materials

The mobile-phone OD tables (TIM) and the Viasat GPS data are proprietary and were used under agreements that do not allow redistribution; requests for access should be addressed to the data providers. MTS loop-detector counts are published by Regione Emilia-Romagna under a CC BY licence (open-data catalogue, dataset "Sistema MTS – Flussi di traffico al quarto d'ora"). Road classes are derived from OpenStreetMap (© OpenStreetMap contributors, ODbL). The road graph and the aggregate results behind the figures and tables are included in the code repository; link-level metrics derived from the GPS data are not distributed. The simulation configurations derive from the mobile-phone OD tables and are not distributed. The simulator `minimocas`, the detector correction and all analysis scripts are available at https://github.com/federicobellisardi/multi-sensor-traffic-reconstruction. Supplementary Section S6 lists the final configuration and the steps that reproduce the results.

Competing interests

C.M. and A.F. are employed by Mindicity, which contributed to the development of the modelling framework. The other authors declare no competing interests.

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Authors' contributions

F.B. designed the study, developed the simulation and analysis pipeline, performed the analyses and wrote the manuscript. C.M. and A.F. contributed to the simulation framework and to the processing of the mobile-phone and detector data. A.B. conceived the modelling approach and supervised the study. All authors read and approved the final manuscript.

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